# Retail Sales & Profit Intelligence Studio

Business Intelligence and Analytics Mini Project · Year 3, Sem V, Computer Engineering · Guide: Dr. Kiran Bhowmick

## In one line

> We take a retail company's sales data, clean it, find where it makes and loses money, predict profit, losses and future sales with machine learning, and show everything on an interactive dashboard.

## Dataset

Official **Sample Superstore** dataset from Tableau: about 10,000 order lines of a retail company in the US and Canada (2023–2026), in three sheets: **Orders**, **Returns** and **People** (regional managers).

| TOTAL SALES | TOTAL PROFIT | MARGIN | ORDERS | CUSTOMERS | RETURNED |
|---|---|---|---|---|---|
| **$2.33M** | **$292K** | **12.6%** | **5,111** | **804** | **5.8%** |

## What we do, step by step

- **1. Data pre-processing** – remove duplicates, fix data types and postal codes, check for invalid values, and join the three sheets into one table.
- **2. Descriptive analytics** – KPIs such as total sales, profit, profit margin, average order value and return rate, broken down by category, region, customer segment and time.
- **3. Data-mining models**
  - **Regression** – predicts the profit of an order (Linear Regression vs Random Forest).
  - **Classification** – predicts whether an order will make a loss; the Decision Tree is about **95% accurate**.
  - **Clustering** – K-Means groups customers into Champions, Loyal, Potential and Low Value, based on how recently, how often and how much they buy (RFM).
  - **Forecasting** – predicts future monthly sales from trend and seasonality.
- **4. Visualization** – 17 charts: trends, bar charts, heatmaps, scatter plots, confusion matrices, customer clusters and the forecast.
- **5. Dashboard** – an interactive black-and-white web app (Streamlit). Filter by date, region, category and segment, and every chart updates.

## Main findings

- **Technology** is the most profitable category; **Tables** and **Bookcases** make a loss.
- Discounts above **20%** are linked to losses (margin falls to -15% and -77% in the higher discount bands).
- The **West** region has the highest profit, but also about **three times more returns** than the other regions.
- Sales peak in **November and December**.

## Model results

| Task | Best model | Result |
|---|---|---|
| Classification (loss or not) | Decision Tree | 95% accuracy, F1 = 0.85 |
| Regression (profit) | Linear Regression | $R^2$ = 0.80 (test split) |
| Forecasting (monthly sales) | Trend + seasonality regression | 17.6% error vs 24.4% baseline |
| Clustering (customers) | K-Means (k = 4) | 4 customer segments |

## Why it is useful

> It helps a manager decide where to cut discounts, which products to fix, which customers to keep, and why one region has so many returns – using data instead of guesswork.

## Tools used

Python · pandas · NumPy · scikit-learn · Matplotlib · Seaborn · Plotly · Streamlit · Jupyter Notebook

Data source: Tableau Public sample data (public.tableau.com/app/learn/sample-data). The company and its transactions are fictitious.